MECHANICAL PROPERTIES OF GEOPOLYMER CONCRETE

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Abstract -

Global warming, One of the main threat of climate change is considered as an environmental challenges of society. Sustainability is solely to induce less carbon dioxide emission, produce more durable concrete with usage of less cement content and to strengthen the structure with longer life span. Geopolymer Concrete (GPC) is an innovative and environment friendly alternative to Ordinary Portland Concrete (OPC) for the use in construction appliance. GPC has attracted prominent attention among the binders and activators. These binders intensify durability performance, sustainability and environmental friendly. The paper provides a complete overview of structural performance of geopolymer concrete and an experimental study is conducted to validate the compressive strength and splitting tensile strength of Geopolymer Concrete.

Keywords – Geopolymer, Sustainability, Durability, Compressive strength, Tensile strength

1. INTRODUCTION

As the demand for concrete usage in construction field increases, it is considered that cement production increases from about 1.5 billion tons to approximately 2.2 billion tons in 2010. The climate change is mainly due to Global warming, it is produced by the emission of greenhouse gases, such as carbon dioxide (CO_2), contributes about 65% of global warming. The cement industry is responsible for these emissions and must be secured by disposal of byproducts. Therefore with regard to global warming, Geopolymer concrete is a promising candidate as an alternative to OPC in the development of alternative binders for the application of concrete structures.

From the literature review, it is noticed that there is no proper design steps for Geopolymer Concrete. The term "Geopolymer" was first introduced by Davidovits in 1978 to describe the binders in chemical composition of a microstructure. As compared to OPC, geopolymers do not form calcium-silicate-hydrates (CSH) gel for strength, but polycondensation process is to acquire strength (Komnitsas, K. A. 2011). The main constituents of geopolymers are source materials and alkaline activator solution. Source materials will be by-products such as fly ash, GGBS (Ground granulated blast furnace slag), rice husk and red mud. GPC is an inorganic polymer to form a significant element of sustainable construction in place of OPC.

Duxson, P., Provis, J. (2005) referred as concrete with alkaline activator solution as a binder produce high strength, durability, stiffness and mechanical properties. The other relative mechanical properties such as density and young's modulus are considerable in structural appliances. The experimental Split tensile strength of Geopolymer Concrete was higher than Ordinary Portland Concrete and Elastic modulus increases as the compressive strength increases (Prem, P. R., Ramachandra Murthy, A., & Bharatkumar, B. H. (2015). As the size of specimen will increase the ratio of compressive failure stress to compressive strength will be decreased (Gonnerman, H. F. 1925).

In a brief, there are intrinsic applications of geopolymer concrete such as bridges, precast structural elements and as well as structural retrofits using geopolymer-fiber composites. GPC approach most advance technology in precast applications due to the relative ease in handling sensitive source materials as alkaline activator solutions. Other than this, there are precast pavers & slabs for paving, bricks and precast pipes (Gourley, J. T., & Johnson, G. B. 2005). As literature talks about characterization of conventional concrete models, they do not differ much when compared to geopolymer concrete. Nowadays, the research gap has been increasing exponentially such that there are many applications of this material. Currently this paper provides an outline of development of geopolymer concrete and

their mechanical properties. The requirement of structural behavior is required for GPC to propel the compressive strength and split tensile strength under uni axial loading. Thus, compressive strength of concrete specimens with homogenous proportions of different sizes and shapes are different.

2. EXPERIMENTAL PROCEDURE

The development of Geopolymer concrete mix by using rich source materials and alkaline activator solutions such as fly ash, GGBS, Sodium hydroxide and Sodium silicate solution respectively. The materials are explained briefly as follows

2.1 Materials

Fly ash

Fly ash is a by-product from burning of coal in electric generating plants and produced in thermal power stations. It is replaced by the natural aggregates and minerals which will reduce considerably the demand and have many applications. Fly ash is very fine material and the particle size varies 10 to 100 micron. The fineness modulus of material is 310 m^2/kg and is spherical glassy shaped. Specific gravity and bulk density of fly ash are 2.21 and 995 Kg/m^3 respectively.

The standard specification used for fly ash is ASTM C618 and based on the chemical composition; these are divided in to two types- Class F and Class C. Here class C tends to high calcium fly ash, based on some physical and chemical properties it is noted that it can acquire more early strength than low calcium based fly ash. From the critical literature, Class C fly ash induce higher compressive strength and less setting time compare to Class F based polymer.

Class F fly ash is used here in the experiment and the chemical composition is shown in the table-1.

GGBS

GGBS is a by-product from the blast furnace which is used to make iron and is main component as an alternative binder. The usage of GGBS is increasing exponentially such that it improves the durability property of concrete. The fineness modulus of material is 425 m^2/kg. Specific gravity and bulk density of GGBS are 2.91 and 1360 Kg/m^3 respectively. It takes slower setting time for casting of super structure and improves the strength. The chemical composition of the material is as shown in the table-1.

Chemical composition	**Fly ash**		**GGBS**
	Class C	**Class F**	
SiO_2	53.1	60.3	35
Al_2O_3	16.5	26.5	12
Fe_2O_3	6.4	7.8	0.2
CaO	16.3	1.3	40
MgO	0.9	0.8	10

Table – 1 Chemical composition of Fly ash and GGBS

Sodium Hydroxide

Sodium Hydroxide also called as caustic soda, is a white crystalline substance. It is easily soluble in water and glycerine. It is very expensive and available in disparate forms as flakes, chips, pellets and as solution. Sodium hydroxide is used as a strong base and concentration can vary from 8 to 16 M for geopolymerisation. Properties of Sodium hydroxide for GPC are as shown in the table-2

Property	Value
Molar mass	40 g/mol
Density	2.1g/cm^3
Melting point	318° C
Boiling point	1390° C
Flash point	Non-flammable

Table -2 Properties of Sodium Hydroxide

Sodium Silicate

Sodium silicate is a crucial constituent of the geopolymer mix. It is also called as water glass, constitute various compounds or combination of sodium silicates varying weight ratio of SiO_2 to Na_2O to be 2:1. As potassium silicate is relatively bit costlier so silicate is most commonly preferred for preparation of AAS solution. These compounds are transparent, powder form and viscous solution in water. They are vaguely soluble and get dissolved by heating with water. The solutions are strongly alkaline. Some of the properties of sodium silicate are shown in the table-3.

Type	Sodium Silicate
Weight ratio	2.2
Molar ratio	2.7
Density	1.59
pH	14.89
Specific gravity	1.6

Table -3 Properties of Sodium silicate

The fine aggregate, coarse aggregate are river sand and crushed aggregate. The specific gravity is 2.47 Kg/m^3 for fine and 2.54 Kg/m^3 for coarse aggregate. Water absorption is 2.85 for fine and 7.21 for coarse type. Water content is also main important in the mix of GPC and major component in the alkaline solution.

2.2 Mixing

Three different types of mix proportions are examined for the engineering properties of Geopolymer Concrete. The alkaline activator solution is made by mixing both the solutions together before 24 hours prior to use. The solutions are strongly alkaline and Sodium hydroxide solution should be dissolved completely in water such that it will be cool for 24 hours to the required concentration. Sodium hydroxide concentration of 5 molar is used here in experiment. At first dry materials i.e., river sand, fly ash, GGBS and crushed aggregate are mixed together, next step is to add the alkaline activator solution to the mixer and mix it well for 3-4 minutes. Extra water is added to the mixture to obtain a good workability. Then the super plasticizer is added slowly and mixes it well for 2-3 minutes. The workability is determined by means of slump test.

2.3 Casting

Here two different sizes of cubes and cylinders are casted i.e.; 100x100x100 mm and 150x150x150 mm cubes, 150x300 mm and 100x200 mm cylinders. Three cubes and cylinders are casted for each mix to determine the average compressive strength and split tensile strength. After the test, the fresh concrete is casted in to the moulds, the specimens are to be compacted into three layers using tamping rod followed by vibration of 10-20 seconds using vibrating table. The casting details are shown in the following figure.

Fig 1 Mixing of materials **Fig 2 Testing of Slump** **Fig 3 Casting of specimens**

3. TEST PROCEDURE

The average of three cubes and cylinders are tested to find out the average compressive strength and split tensile strength at the age of 28 days.

3.1 Compressive Strength

The Compressive strength is carried out on HT-8391PC (Computer-Control Servo Hydraulic Concrete Testing-machine) to determine the strength of concrete specimen at 28 days. Standard cube specimens (100x100x100) mm and (150x150x150) mm are tested under 2500 kN capacity are placed at the centre, the load is applied and ultimate strength is acquired till the specimen fail. The values are recorded and the average compressive strength is calculated. The maximum load per unit area gives the compressive strength.

3.2 Split Tensile Strength

The Tensile strength is carried out on UTM (Universal testing machine) of 1000 kN capacity to determine the strength at 28 days. Standard cylinder specimens (100x200) mm and (150x300) mm are marked at each end of the specimen and placed between two wooden strips, such that the load is placed exactly at the centre. The loading rate is kept constant until it fails and average tensile strength is calculated.

4. RESULTS & DISCUSSION

The failure patterns of cube and cylinder specimens are shown in Fig 4 and 5. The results of compressive strength and split tensile strength of specimens are shown in the table 4 and 5.

Fig 4 Compressive Strength **Fig 5 Split Tensile Strength**

Specimen type	No.	MIX 1 (N/mm^2)	MIX 2 (N/mm^2)	MIX 3 (N/mm^2)
Cubes (100x100x100)	1	49.61	30.75	60.30
	2	55.52	33.69	60.46
	3	54.33	30.15	55.04
Average Strength		**53.15**	**31.53**	**55.04**
(150x150x150)	1	57.82	35.89	63.04
	2	61.81	32.30	67.49
	3	57.36	33.58	70.41
Average Strength		**59.01**	**33.92**	**66.98**

Table -4 Compressive Strength of Specimens at 28 days

Specimen type	MIX 1 (N/mm^2)	MIX 2 (N/mm^2)	MIX 3 (N/mm^2)
Cylinders (100x200)	3.57	3.24	3.31
(150x300)	4.09	2.76	4.27

Table -5 Split Tensile Strength of Specimens at 28 days

- The compressive strength is observed at 28 days and the peak point is noted as the maximum compressive strength of 53.15, 31.53, 55.04 N/mm^2 of mix 1, 2 and 3 respectively for 100 mm cube specimen.
- At the same instant the peak point as 59.01, 33.92, 66.98 N/mm^2 of mixes for 150 mm cube specimen respectively.
- Therefore the maximum compressive strength of geopolymer concrete is observed and it tends to be more than 70 Mpa.
- Split tensile strength of GPC increases with more availability of Sodium silicate ratio in the mix and the range of Split tensile strength of concrete vary from 3 to 5 N/mm^2.
- As seen in the graph compressive strength goes on increasing as the size of specimen increases at the age of 28 days.

The following graph shows compressive strength of cube specimens of different sizes at 28 days.

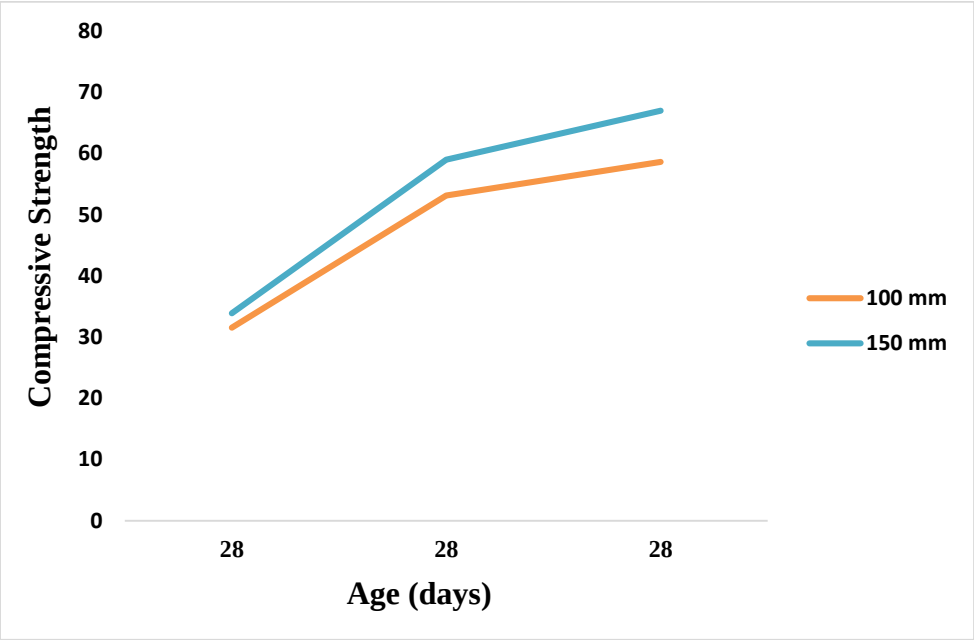

Fig 6 Compressive Strength at 28 days

5. CONCLUSIONS

From the results acquired in the experimental study, ultimately the conclusions are as follows:

- GPC specimens signify higher compressive strength than compared to OPC for which gap between the compressive strength is slightly varying as the specimen size increases.
- The crucial point is the cost of alkaline activator solution which is used for GPC mix that shape less productivity in cost.
- Applications of GPC are quite limited and show better durability properties of structure than OPC.
- The other important thing is to control lack of using super plasticizer to achieve workability of GPC and to recommend the codes for GPC at material and structural purpose in practice.

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